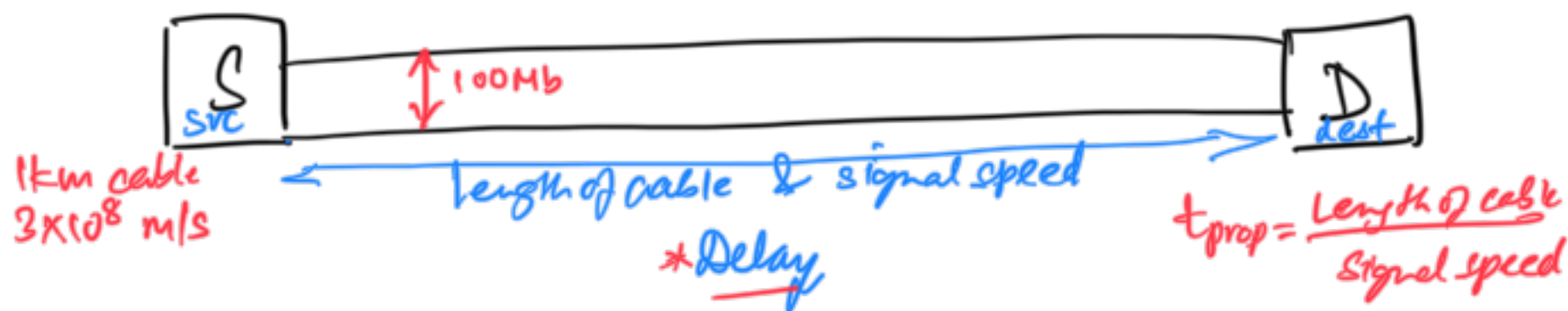


Lec 04: Perf modelling of computer networks.

Two main attributes to measure network performance 16/8/23

* Throughput / Bandwidth = $\frac{\text{No. of bits that can be sent}}{\text{Total time reqd to send}}$



t_{prop} : Time reqd for '1' bit to travel from one end to another

* What are conditions that determine throughput?

① Bandwidth or Data rate $\Rightarrow$ Bottleneck bandwidth

② Protocol Frame transmission time, t_f

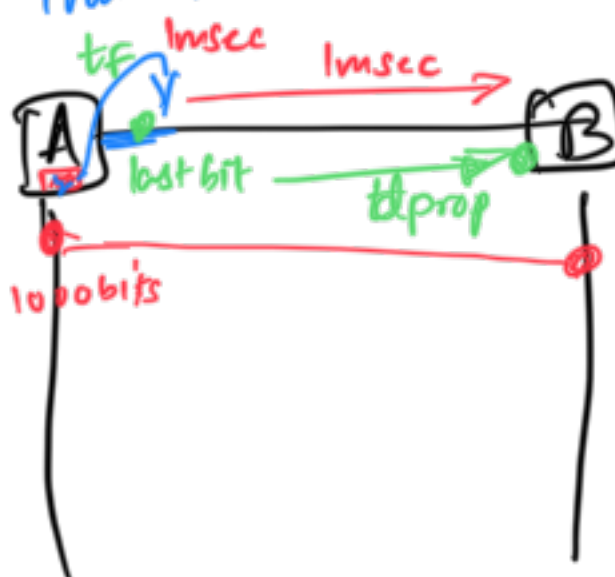
E.G.

$t_{prop} = 1\text{ms}$

$L = 1000\text{bits}$ Packet Size

$R = 1\text{Mbps}$

Data rate / BW



1Mbps

Scenario 1: no Ack reqd

$T_{pt} =$

$10^6 \text{ bits} \dots 1\text{sec}$

$10^3 \text{ bits} \dots t_f$

$t_f = \frac{L}{R} = \frac{10^3}{10^6} = 1\text{msec}$

define

End-to-end delay = $t_f + t_{prop}$

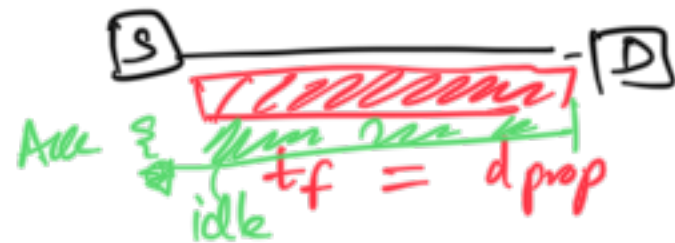
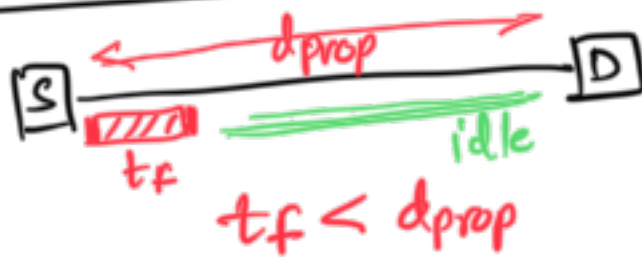
$= 1 + 1$

$= 2\text{msec}$

Throughput = $\frac{L}{\text{define}}$ = $\frac{10^3 \text{ bits}}{2\text{msec}}$

$= 500 \text{ Kbps}$

Scenario 2: Wait for an Ack after every data packet

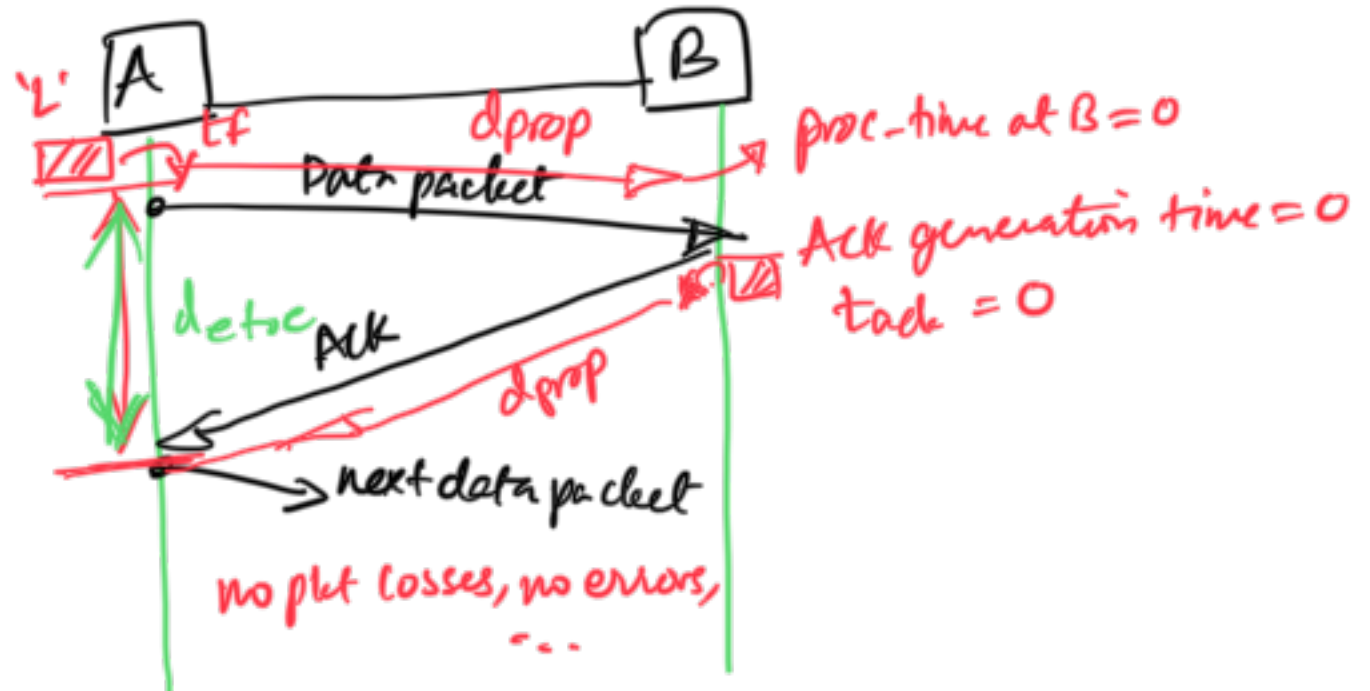


E-G.

$L = 1000 \text{ bits}$

$R = 1 \text{ Mbps}$

$d_{prop} = 1 \text{ ms}$



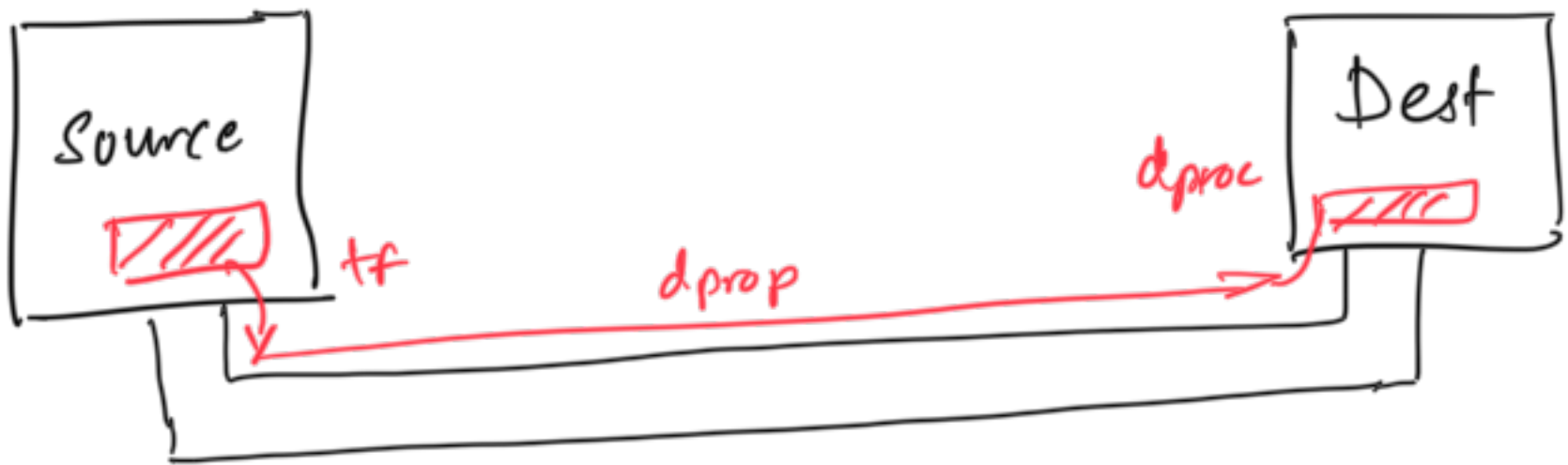
$$\begin{aligned} d_{etoe} &= t_f + d_{prop} + d_{prop} \\ &= 1 + 1 + 1 \\ &= \underline{\underline{3 \text{ msec}}} \end{aligned}$$

$$\text{Throughput} = \frac{L}{d_{etoe}} = \frac{1000 \text{ bits}}{3 \text{ msec}} = \underline{\underline{333 \text{ kbps}}}$$

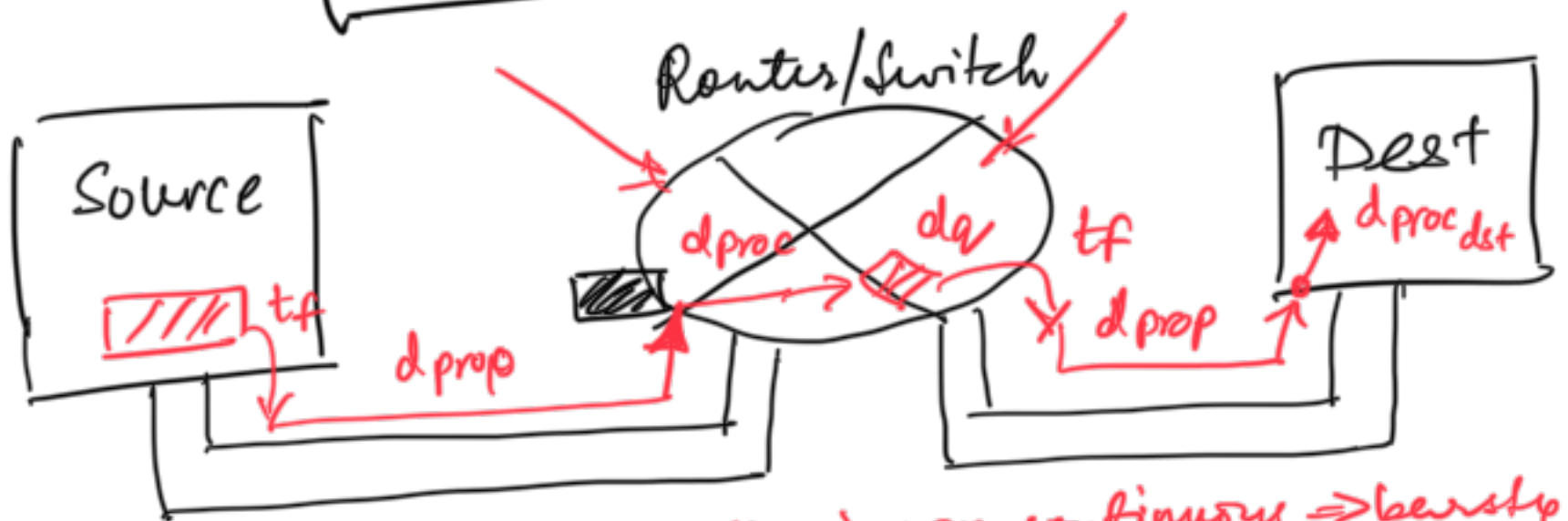
③ Delays $\Rightarrow$ due to packet errors, retransmissions, queuing, etc.

→ Types of delays.

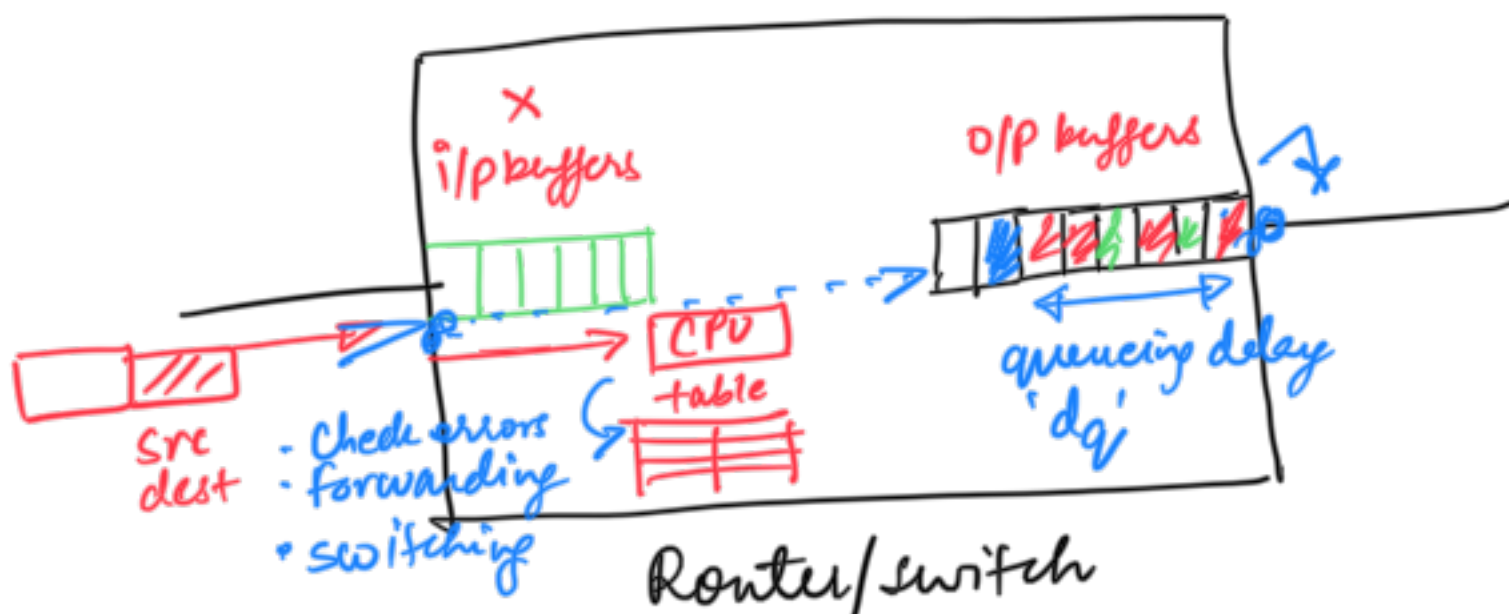
1. **Transmission delay** d_{trans} or t_f
 - Time required to place the data from the device buffer to the channel.
 - $t_f = \frac{L}{R}$ → data size
→ data rate of channel
2. **Propagation delay** d_{prop}
 - Time reqd for 1 bit to propagate from one end to the other.
 - $d_{prop} = \frac{\text{Distance (or cable length)}}{\text{Signal speed}}$
3. **Processing delay** d_{proc}
 - Time need to process the data before forwarding/accepting it.
 - includes time reqd to check errors, decide the next hop via table lookup, etc. *not significant since runs in hardware at wire speed.*
4. **Queueing delay** d_q
 - Time spent by the data packet in a router's (or switch) queue.
 - non-deterministic delay
 - ↳ not possible to determine the instantaneous queue size in an Internet router
 - ↳ Data traffic in Internet is non-contiguous or bursty in nature.



Store & Forward Switching



data traffic is non-continuous $\Rightarrow$ bursty



Problem 1

$$L = 125 \text{ KB} = 125 \times 8 \times 10^3 \text{ bits} = 10^6 \text{ bits}$$

$$R = 1 \text{ Mbps}$$

$$d_{\text{tot}} = t_f + d_{\text{prop}}$$

$$= \frac{10^6}{10^6} + 0$$

$$d_{\text{tot}} = 1 \text{ sec}$$

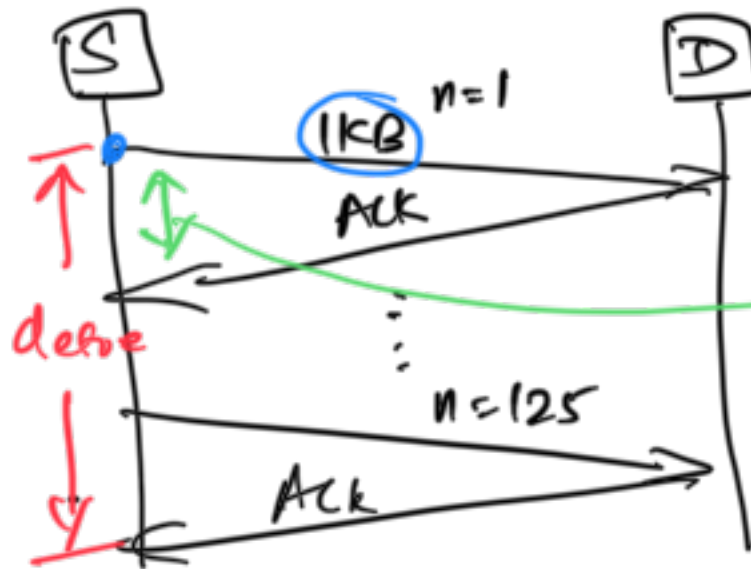
Problem 2.

$$\text{File} = 125 \text{ KB}$$

$$L = 1 \text{ KB}$$

$$\# \text{ pkts} = 125$$

$$d_{\text{prop}} = 20 \text{ ms}$$



$$t_f = \frac{L}{R} = \frac{8000}{10^6} = 8 \text{ ms}$$

$$t_f + \text{back} + d_{\text{prop}} + d_{\text{prop}} = 8 + 20 + 20 = 48 \text{ msec}$$

$$d_{\text{tot}} = 48 \text{ ms} \times 125$$

$$d_{\text{tot}} = 6 \text{ sec}$$

$$\text{Throughput} = \frac{125 \text{ KB}}{6 \text{ sec}}$$

$$\frac{1 \text{ KB}}{48 \text{ msec}} =$$

